Midterm 2 — Sample 2

DATA 412/612

Practice exam, questions only. Put your answers in an R Markdown document, and make sure code is visible when you knit (`knitr::opts_chunk$set(echo = TRUE)` in the first chunk). Use the tidyverse rather than base R where you can. You may use your notes, the R help files, and the internet. Do not discuss the exam with anyone else.

This sample covers the same ground as Sample 1 (joins, strings, and tidying), and it also uses factors and dates.

1. **Library visits** (10 pts).

   `branches`: `branch` (name), `city`, `seats`.

   `visits`: `patron`, `branch`, `check_in` (date and time, stored as text), `minutes` (length of the visit).

```
library(tidyverse)

branches <- tribble(
  ~branch,    ~city, ~seats,
  "Shaw",     "DC",  80,
  "Petworth", "DC",  45,
  "Rosslyn",  "VA",  60
)

visits <- tribble(
  ~patron, ~branch,    ~check_in,             ~minutes,
  "Ari",   "Shaw",     "Mar 03, 2024 9:15am",  40,
  "Bea",   "Rosslyn",  "Mar 03, 2024 2:05pm",  15,
  "Cal",   "Petworth", "Mar 04, 2024 11:00am", 90,
  "Dee",   "Shaw",     "Mar 04, 2024 6:40pm",  25
)
```

   1. (3 pts) Use `parse_datetime()` to parse `check_in`, and assign the result back to `visits`.
   2. (3 pts) Add `city` and `seats` from `branches` onto `visits`.
   3. (4 pts) Add a column `weekday`: an ordered factor for the weekday of `check_in`, with levels Monday through Sunday in that order.

2. **Strings** (10 pts). Use only functions from the tidyverse. Your code should work for strings of this shape in general, not only for the rows below.

```
notes <- tribble(
  ~ratio, ~words,
  "3/4",  "Undo",
  "5/2",  "into",
  "1/8",  "pretty",
  "9/9",  "ready",
  "2/5",  "inner",
  "7/10", "are.",
  "4/4",  "Lost"
)
```

1. (4 pts) Convert `ratio` from a character vector of fractions to a numeric vector of decimal values. For example, `"3/4"` should become `0.75`. No `stringr` functions and no regular expressions.

2. (4 pts) Use a single regular expression to find every word in `words` that begins with `un` or `in`, ignoring case, and replace that prefix with `re`. So `Undo` becomes `redo`, and `into` becomes `reto`.

3. (2 pts) **Challenge.** Move the first character of each word to the end, then append `ay`. Examples: `study` becomes `tudysay`, and `aloe` becomes `loeaay`. Build that version of each string in `words`.

   For a little extra credit: keep an initial capital as a capital on the new first letter (`John` becomes `Ohnjay`), and keep a trailing period in place (`goodbye.` becomes `oodbyegay.`).

3. **Tidying a score file** (20 pts). The file `survey_scores.csv` is on the course page under Data. Each row is one person. The condition columns are `respondent`, `city`, `year`, and `arm`.

   `arm` is a string. It holds either one letter in parentheses, such as `(A)`, or two letters in parentheses separated by a comma and a space, such as `(A, B)`.

   The other columns are scores. The name is a time, an underscore, and a subject: `pre_math`, `pre_read`, `post_math`, `post_read`, `follow_math`, and `follow_read`. The column `drop_` is a score with no subject.

   For each part, print the first six rows with `head()`. The only non-tidyverse functions you may use in this question are `head()` and `c()`.

   1. (2 pts) Use `readr` to load the file, and view the first six rows.

   2. (4 pts) Reshape the data so that, besides the four condition columns, one column holds the score-column name and another column holds the score.

   3. (3 pts) Split that name into a time column and a subject column. For `drop_`, the subject should be the empty string `""`.

   4. (5 pts) Clean `arm`:

      - Remove every parenthesis.
      - Replace the first letter with that same letter followed by an underscore.
      - Remove every comma and the space after it.

      For example, `(A, B)` becomes `A_B`, and `(A)` becomes `A_`.

   5. (3 pts) Using the cleaned `arm`, make two columns: the first letter and the second letter. If `arm` had only one letter, the second is `NA`. Keep both columns as character, not numbers.

   6. (3 pts) Make an appropriate plot of how time and subject relate to the score. Leave out the `drop` rows, which have no subject.